

COMPREHENSIVE SRS DOCUMENT – BLUE NILE COFFEE PLATFORM

AI-Enhanced E-Commerce System Specification

1. INTRODUCTION

This Software Requirements Specification (SRS) outlines the full technical framework for the Blue Nile Coffee AI-driven e-commerce platform. It consolidates all system modules including personalization, recommendations, predictive analytics, automation, workflow tools, influencer content, and analytics integrations.

2. SYSTEM OVERVIEW

The platform uses a decoupled architecture: • Frontend: React / Vue / Next.js • Backend: Laravel API-driven framework • AI Layer: External APIs (Clerk.io, Pinecone, OpenAI, Mixpanel) • Analytics Layer: Mixpanel, TikTok Pixel, Meta Pixel, Google Analytics

3. AI PERSONALIZATION & RECOMMENDATION ENGINE

3.1 Clerk.io Integration

Website: <https://www.clerk.io>

Clerk.io provides real-time product recommendations, personalized search, upsell and cross-sell automation, email recommendations, and dynamic product ranking.

Pricing Tiers:

- Starter: \$99–\$149/month (small stores)
- Professional: \$149–\$249/month (mid-sized stores)
- Enterprise: \$349+/month

3.2 Pinecone + OpenAI Embeddings

Pinecone: <https://www.pinecone.io>

OpenAI: <https://platform.openai.com>

This setup stores vector embeddings representing product metadata and customer behaviour. It enables similarity search and intelligent recommendation flows.

Pinecone Pricing:

- Free Tier
- Standard Pods: ~\$0.096–\$0.29 per GB-hour

OpenAI Embeddings:

- Approx. \$0.02 per 1M tokens

4. PREDICTIVE ANALYTICS & GROWTH INSIGHTS

Mixpanel Analytics

Website: <https://mixpanel.com>

Mixpanel provides behavioural analytics including funnels, churn prediction, retention tracking, repeat purchasing analysis, and customer cohort breakdowns.

Pricing:

- Free: Up to 20M monthly events
- Growth Plan: ~\$20–\$35/month
- Enterprise: Custom pricing

5. AI CONTENT & CHAT AUTOMATION

OpenAI GPT-4/GPT-5 enables automated support chat, description writing, email automation, influencer script generation, and dynamic product content.

6. WORKFLOW AUTOMATION

Tools: Zapier (<https://zapier.com>), Make (<https://make.com>)

Used for CRM automation, order updates, customer notifications, internal workflow triggers, and recurring marketing automations.

7. AI VIDEO & INFLUENCER TOOLING

AI-driven platforms such as D-ID, Synthesia, and Runway are used to generate influencer-style UGC videos for ads, social media, and promotional material.

8. SYSTEM ARCHITECTURE SUMMARY

1. User accesses frontend application 2. Frontend communicates with Laravel backend via REST APIs 3. Backend integrates with AI APIs for recommendations and analytics 4. Behaviour analytics captured through Mixpanel and Pixels 5. Automation layer handles workflows

9. ESTIMATED MONTHLY AI OPERATING COST

• Small e-commerce: ~\$120/month • Mid-sized: \$250–\$500/month • Large enterprise: \$800+/month based on API volume

10. CONCLUSION

This comprehensive SRS defines the AI-driven architecture required for a scalable and automated e-commerce ecosystem for Blue Nile Coffee and can be applied to any related dropshipping or multi-brand coffee platform.